

Ejercicios de códigos.

1. Invertir un arreglo de enteros

```
#include <stdio.h>

int main() {
    int a[] = {1, 2, 3, 4, 5}, b = 0, c = 4, d;
    while (b < c) {
        d = a[b];
        a[b] = a[c];
        a[c] = d;
        b++;
        c--;
    }
    for (b = 0; b < 5; b++) {
        printf("%d ", a[b]);
    }
    return 0;
}
```

2. Contar la longitud de una cadena sin `strlen()`

```
#include <stdio.h>

int main() {
    char a[] = "hola";
    int b = 0;
    while (a[b] != '\0') {
        b++;
    }
    printf("%d\n", b);
    return 0;
}
```

3. Sumar los elementos de un arreglo

```
#include <stdio.h>

int main() {
    int a[] = {1, 2, 3, 4}, b = 0, c = 0;
    for (b = 0; b < 4; b++) {
        c += a[b];
    }
    printf("%d\n", c);
    return 0;
}
```

4. Encontrar el número más grande en un arreglo

```
#include <stdio.h>

int main() {
    int a[] = {3, 7, 2, 9}, b, c = a[0];
    for (b = 1; b < 4; b++) {
        if (a[b] > c) {
            c = a[b];
        }
    }
    printf("%d\n", c);
    return 0;
}
```

5. Contar cuántas veces aparece un número en un arreglo

```
#include <stdio.h>

int main() {
    int a[] = {1, 2, 3, 2, 4}, b = 2, c = 0, d;
    for (d = 0; d < 5; d++) {
        if (a[d] == b) {
            c++;
        }
    }
    printf("%d\n", c);
    return 0;
}
```

6. Copiar una cadena en otra

```
#include <stdio.h>

int main() {
    char a[] = "texto", b[20];
    int i = 0;
    while (a[i] != '\0') {
        b[i] = a[i];
        i++;
    }
    b[i] = '\0';
    printf("%s\n", b);
    return 0;
}
```

7. Comparar dos cadenas carácter por carácter

```
c
CopiarEditar
#include <stdio.h>

int main() {
    char a[] = "hola", b[] = "hola";
    int i = 0, c = 1;
    while (a[i] != '\0' || b[i] != '\0') {
        if (a[i] != b[i]) {
            c = 0;
            break;
        }
        i++;
    }
    if (c == 1) {
        printf("iguales\n");
    } else {
        printf("diferentes\n");
    }
    return 0;
}
```

8. Convertir minúsculas a mayúsculas

```
#include <stdio.h>

int main() {
    char a[] = "hola", b;
    int i = 0;
    while (a[i] != '\0') {
        b = a[i];
        if (b >= 'a' && b <= 'z') {
            b = b - 32;
        }
        printf("%c", b);
        i++;
    }
    return 0;
}
```

9. Imprimir solo los números en una cadena

```
#include <stdio.h>

int main() {
    char a[] = "Hola123";
    int i = 0;
    while (a[i] != '\0') {
        if (a[i] >= '0' && a[i] <= '9') {
            printf("%c", a[i]);
        }
        i++;
    }
    return 0;
}
```

10. Buscar subcadena en una cadena

```
#include <stdio.h>

int main() {
    char a[] = "hola mundo", b[] = "mundo";
    int i = 0, j, k, c = 0;
    while (a[i] != '\0') {
        if (a[i] == b[0]) {
            k = i;
            j = 0;
            while (b[j] != '\0' && a[k] == b[j]) {
                j++;
                k++;
            }
            if (b[j] == '\0') {
                c = 1;
                break;
            }
        }
        i++;
    }
    if (c == 1) {
        printf("si\n");
    } else {
        printf("no\n");
    }
    return 0;
}
```

11. Intercambiar dos arreglos

```
#include <stdio.h>

int main() {
    int a[] = {1, 2, 3}, b[] = {4, 5, 6}, c, i;
    for (i = 0; i < 3; i++) {
        c = a[i];
        a[i] = b[i];
        b[i] = c;
    }
    for (i = 0; i < 3; i++) {
        printf("%d ", a[i]);
    }
    return 0;
}
```

12. Promedio de un arreglo

```
#include <stdio.h>

int main() {
    int a[] = {1, 2, 3, 4}, b, s = 0;
    for (b = 0; b < 4; b++) {
        s += a[b];
    }
    printf("%.2f\n", s / 4.0);
    return 0;
}
```

13. Reemplazar un carácter en una cadena

```
#include <stdio.h>

int main() {
    char a[] = "casa";
    int i = 0;
    while (a[i] != '\0') {
        if (a[i] == 'a') {
            a[i] = 'x';
        }
        i++;
    }
    printf("%s\n", a);
    return 0;
}
```

14. Imprimir solo mayúsculas

```
#include <stdio.h>

int main() {
    char a[] = "Hola Mundo";
    int i = 0;
    while (a[i] != '\0') {
        if (a[i] >= 'A' && a[i] <= 'Z') {
            printf("%c", a[i]);
        }
        i++;
    }
    return 0;
}
```

15. Contar una letra específica

```
#include <stdio.h>

int main() {
    char a[] = "repetir", b = 'e';
    int i = 0, c = 0;
    while (a[i] != '\0') {
        if (a[i] == b) {
            c++;
        }
        i++;
    }
    printf("%d\n", c);
    return 0;
}
```

16. Ver si una cadena es palíndromo

```
#include <stdio.h>

int main() {
    char a[] = "oso";
    int i = 0, j = 0, c = 1;
    while (a[j] != '\0') {
        j++;
    }
    j--;
    while (i < j) {
        if (a[i] != a[j]) {
            c = 0;
            break;
        }
        i++;
        j--;
    }
    if (c == 1) {
        printf("si\n");
    } else {
        printf("no\n");
    }
    return 0;
}
```

17. Eliminar espacios

```
#include <stdio.h>

int main() {
    char a[] = "hola mundo";
    int i = 0;
    while (a[i] != '\0') {
        if (a[i] != ' ') {
            printf("%c", a[i]);
        }
        i++;
    }
    return 0;
}
```

18. Ordenar arreglo (burbuja)

```
#include <stdio.h>

int main() {
    int a[] = {3, 1, 4, 2}, i, j, t;
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3 - i; j++) {
            if (a[j] > a[j + 1]) {
                t = a[j];
                a[j] = a[j + 1];
                a[j + 1] = t;
            }
        }
    }
    for (i = 0; i < 4; i++) {
        printf("%d ", a[i]);
    }
    return 0;
}
```

19. Concatenar dos cadenas carácter por carácter

```
#include <stdio.h>

int main() {
    char a[] = "Hola", b[] = "Mundo", c[20];
    int i = 0, j = 0;
    while (a[i] != '\0') {
        c[i] = a[i];
        i++;
    }
    while (b[j] != '\0') {
        c[i] = b[j];
        i++;
        j++;
    }
    c[i] = '\0';
    printf("%s\n", c);
    return 0;
}
```

20. Imprimir solo letras

```
#include <stdio.h>

int main() {
    char a[] = "123abc#&!";
    int i = 0;
    while (a[i] != '\0') {
        if ((a[i] >= 'a' && a[i] <= 'z') || (a[i] >= 'A' && a[i] <= 'Z'))
        {
            printf("%c", a[i]);
        }
        i++;
    }
    return 0;
}
```